La Scuola d'Italia / Dr. Huson / IB Math
23 April 2026

First & Last name:

7.1 Classwork: Introduction to Logarithms (no calculator)

1. Evaluate each of the following.

   (a) $\log_4 \dfrac{1}{64}$

   (b) $\log_{25} 5$

   (c) $\log_2 \sqrt{32}$

   (d) $\log_5 125$

   (e) $\log_{16} \dfrac{1}{4}$

   (f) $\log_3 \sqrt{81}$

2. Let $c = \log_3 m$, where $m > 0$. In each question, write the expression in terms of c.

   (a) $\log_3 m^2$

   (b) $\log_3(27m)$

   (c) $\log_9 m$

   Let $a = \log_2 m$, where $m > 0$. Write the expression in terms of a.

   (d) $\log_2 m^3$

   (e) $\log_2(8m)$

   (f) $\log_4 m$

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3. Use the laws of logarithms.

   (a) Express each in terms of $\log x$ and $\log y$.

      (i) $\log(100x)$

      (ii) $\log\left(\dfrac{x^2}{y}\right)$

      (iii) $\log\sqrt{xy^3}$ [6]

   (b) Write each expression as a single logarithm.

      (i) $\log_2 x + \log_2 y$

      (ii) $2\log_7 a - \log_7 b$

      (iii) $\dfrac{1}{2}\ln p + \ln q$

4. Solve each equation. Give exact answers.

   (a) $\log_2(x-3) = \log_4(x+9)$

   (b) $\log_3 x = 2$

   (c) $\log_5(2x) = 1$

   (d) $\log_2(x+1) + \log_2(x-1) = 3$